ORBITAL MECHANICS OF PLANETS — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Orbital Mechanics of Planets

Question 1

10.0 points

Examine the diagram below depicting a planet's elliptical orbit around the sun. Using Kepler's second law, describe how the area swept out by a line from the planet to the sun varies over time. Explain how the speed of the planet changes as it moves along different segments of its orbit and why.

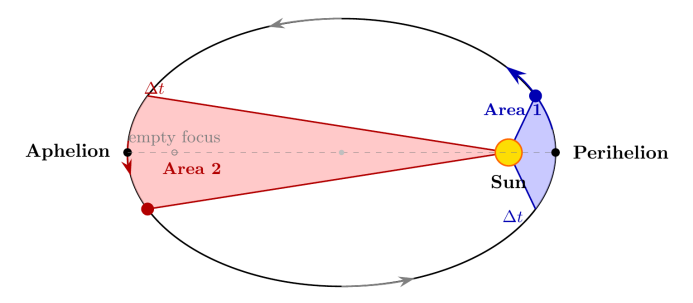

Fig. 1. Diagram of a planet's elliptical orbit around the sun. The sun is located at one focus of the ellipse. Two positions of the planet are shown along the orbit: one closer to the sun and one farther away. Lines are drawn from the sun to the planet at these two positions. Shaded areas are shown between these lines and the orbit, representing the area swept out over equal time intervals. Arrows indicate the direction of the planet's motion, showing faster movement when closer to the sun and slower movement when farther away.

Question 2 10.0 points

Refer to the diagram below showing a spacecraft executing a Hohmann transfer orbit between two circular orbits around Earth. Calculate the velocity changes (Δv) required at each segment of the transfer. Provide a detailed step-by-step explanation of the calculation.

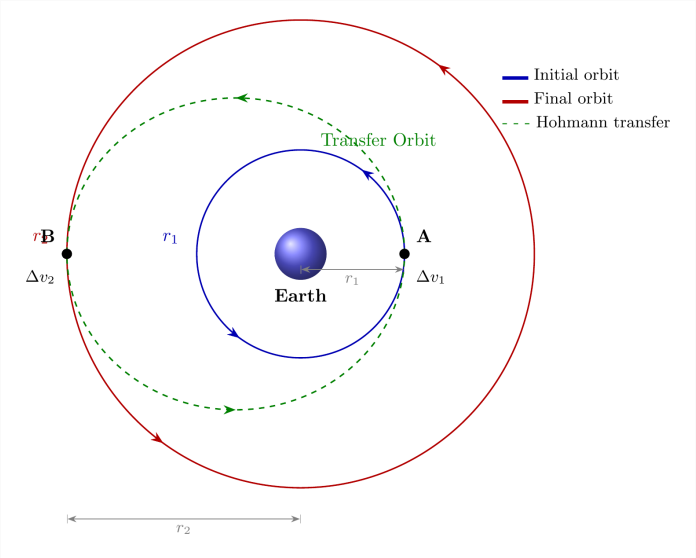

Fig. 2. REQUIRED LABELS (must appear visibly in the diagram): Δv
A diagram showing a Hohmann transfer orbit between two circular orbits around Earth. The diagram should include: 1) Earth at the center, 2) an inner circular orbit and an outer circular orbit, 3) the elliptical transfer orbit connecting the two circular orbits, 4) points labeled where velocity changes (Δv) occur at the perigee and apogee of the transfer orbit. The orbits should be labeled with their respective radii, and the transfer orbit should be shown as a dashed ellipse for clarity.

Question 3

10.0 points

Analyze the diagram below illustrating the trajectory of a space probe performing a gravitational slingshot maneuver around a planet. Explain how the slingshot alters the probe's trajectory and velocity. Discuss the conservation laws applicable in this maneuver, considering <eq_q_1> and <eq_q_2>.

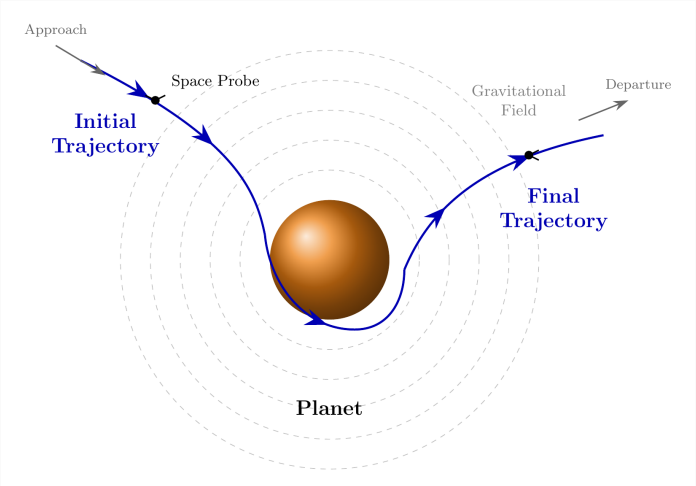

Fig. 3. Diagram illustrating the trajectory of a space probe performing a gravitational slingshot maneuver around a planet. The diagram shows the planet at the center with its gravitational field represented by concentric circles. The space probe's initial trajectory approaches the planet, curves around it due to gravitational attraction, and exits along a new trajectory. Arrows indicate the direction of the probe's velocity before and after the maneuver. Labels include 'Initial Trajectory', 'Final Trajectory', 'Planet', and 'Gravitational Field'.

Question 4

10.0 points

Analyze the diagram below that illustrates the five Lagrange points (L1-L5) in the Earth-Sun system. Describe the stability of each point and how satellites can utilize these positions.

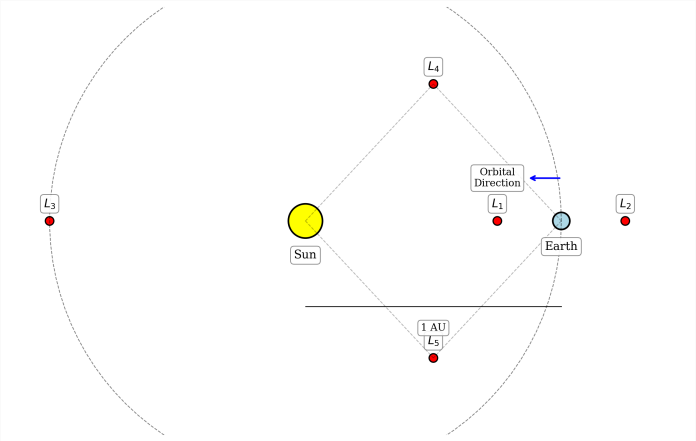

Fig. 4. Diagram of the Earth-Sun system showing the five Lagrange points (L1, L2, L3, L4, L5). The Sun is on the left, the Earth is on the right, and the Lagrange points are labeled in their respective positions: L1 between the Earth and Sun, L2 beyond the Earth, L3 on the opposite side of the Sun from the Earth, L4 and L5 forming an equilateral triangle with the Earth and Sun. Include labels for the Earth, Sun, and each Lagrange point.

Question 5 10.0 points

Based on the diagram below detailing Mercury's orbit, explain the phenomenon of orbital precession and its causes, including both Newtonian mechanics and general relativity contributions, with equations <eq_q_1> and <eq_q_2> resolved.

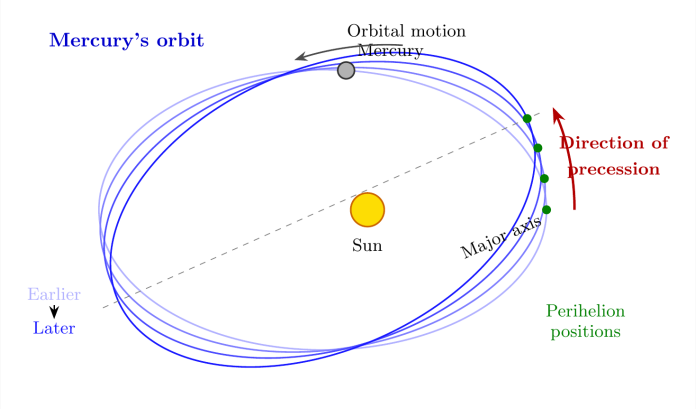

Fig. 5. Diagram showing Mercury's elliptical orbit around the Sun with an exaggerated precession effect. The orbit should be an ellipse with the Sun at one focus. Indicate the direction of precession with an arrow along the orbit. Label the major axis of the ellipse and show the angle of precession. Include labels for Newtonian mechanics and general relativity contributions to the precession. The diagram should help visualize how the orbit shifts over time due to these effects. IMPORTANT: The diagram is missing required labels. The diagram is missing a required label for. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram description explicitly requires a label for the. Ensure ALL structural components mentioned in the description are clearly labeled.